



Lab 13
Course: Networks System Design
Name: Do Davin
Student ID: P20230018
Instructor: Mr. Kuy Movsun

Part 1: Infrastructure Mode & Connectivity

Physical Config GUI Attributes

Wireless-N Broadband Router

Wireless Setup Wireless Security Access Restrictions Applications & Gaming Admin

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter

Wireless Security

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: Wireless@Admin2026

Key Renewal: 3600 seconds

Wireless Router0

Physical Config GUI Attributes

Wireless-N Broadband Router

Wireless Setup Wireless Security Access Restrictions Applications & Gaming Administration

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter

Basic Wireless Settings

Network Mode: Mixed

Network Name (SSID): NET_LAB_SECURE

Radio Band: Auto

Wide Channel: Auto

Standard Channel: 1 - 2.412GHz

SSID Broadcast: ☒ Enabled ☐ Disabled

Laptop0

Physical Config Desktop Programming Attributes

IP Configuration

Interface Wireless0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address192.168.0.100

Subnet Mask255.255.255.0

Default Gateway192.168.0.1

DNS Server0.0.0.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

Link Local AddressFE80:2D0:D3FF:FE70:6C74

Default Gateway

DNS Server

Laptop1

Physical Config Desktop Programming Attributes

IP Configuration

Interface Wireless0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address192.168.0.101

Subnet Mask255.255.255.0

Default Gateway192.168.0.1

DNS Server0.0.0.0

IPv6 Configuration

☒ Automatic

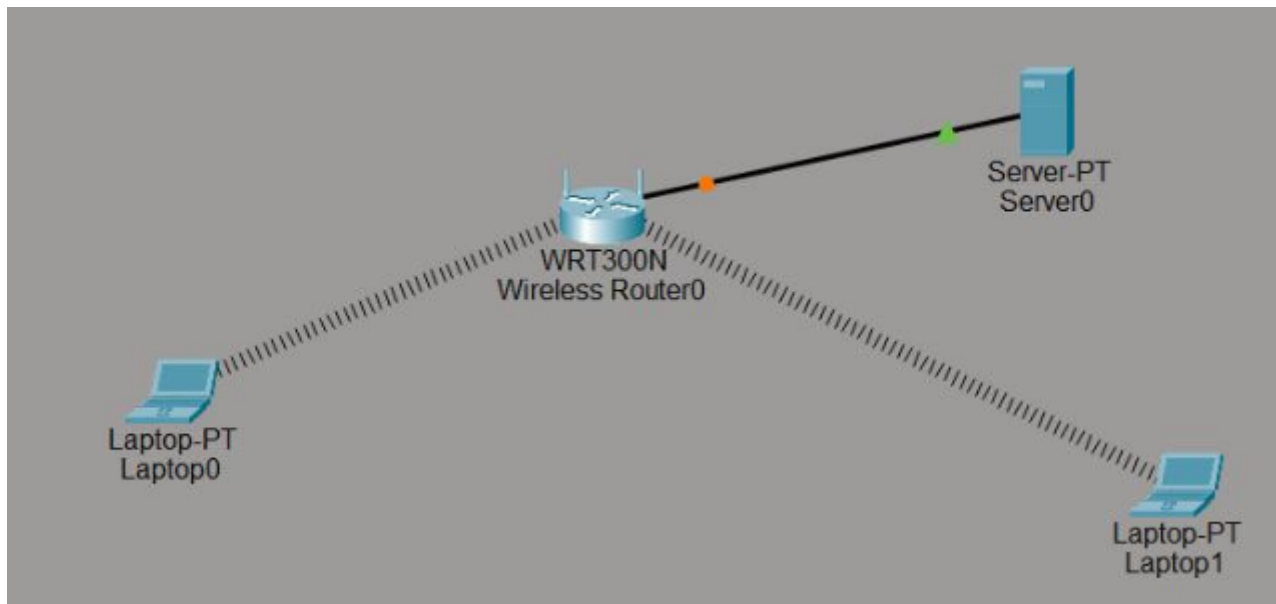
☐ Static

IPv6 Address

Link Local AddressFE80:2D0:58FF:FEED:CB6B

Default Gateway

DNS Server



```
C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=33ms TTL=128
Reply from 192.168.0.100: bytes=32 time=14ms TTL=128
Reply from 192.168.0.100: bytes=32 time=20ms TTL=128
Reply from 192.168.0.100: bytes=32 time=20ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 33ms, Average = 21ms
```

```
Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=3ms TTL=128
Reply from 192.168.0.100: bytes=32 time=4ms TTL=128
Reply from 192.168.0.100: bytes=32 time=3ms TTL=128
Reply from 192.168.0.100: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>
```

Part 2: Analyzing Wireless Management Frames

Event List					
Vis.	Time(sec)	Last Device	At Device		Type
	0.000	--	Laptop0		ICMP
	0.000	--	Laptop0		ARP
	0.001	Laptop0	Wireless Router0		ARP
	0.002	Wireless Router0	Server0		ARP
	0.005	--	Wireless Router0		ARP
	0.006	Wireless Router0	Laptop1		ARP
	0.006	Wireless Router0	Laptop0		ARP
	0.011	--	Wireless Router0		ARP
	0.012	Wireless Router0	Laptop1		ARP
	0.012	Wireless Router0	Laptop0		ARP
	0.012	--	Laptop0		ICMP
	0.015	--	Laptop0		ICMP
	0.016	Laptop0	Wireless Router0		ICMP
	0.021	--	Wireless Router0		ICMP
	0.022	Wireless Router0	Laptop1		ICMP
	0.022	Wireless Router0	Laptop0		ICMP
	0.027	--	Wireless Router0		ICMP
	0.028	Wireless Router0	Laptop1		ICMP

PDU Information at Device: Wireless Router0

OSI Model

Inbound PDU Details

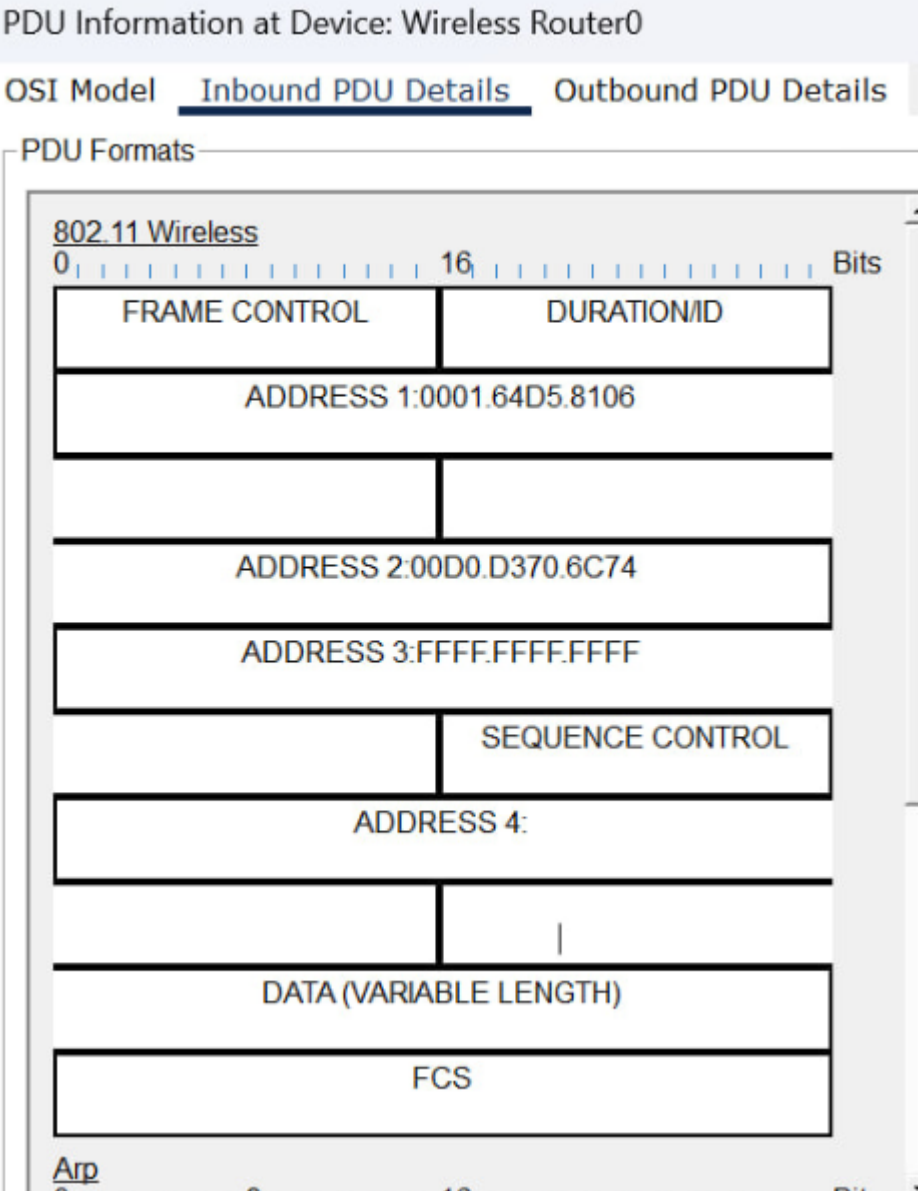
Outbound PDU Details

At Device: Wireless Router0

Source: Laptop0

Destination: Broadcast

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer 2: Wireless ARP Packet Src. IP: 192.168.0.98, Dest. IP: 192.168.0.1	Layer 2: Ethernet II Header 00D0.D370.6C74 >> FFFF.FFFF.FFFF ARP Packet Src. IP: 192.168.0.98, Dest. IP: 192.168.0.1
Layer 1: Port Wireless	Layer 1: Port(s): Ethernet 1



Part 3: Channel Management and Frequency Reuse

Access Point0

Physical Config Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID NET_LAB_SECURE

2.4 GHz Channel 11

Coverage Range (meters) 140.00

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

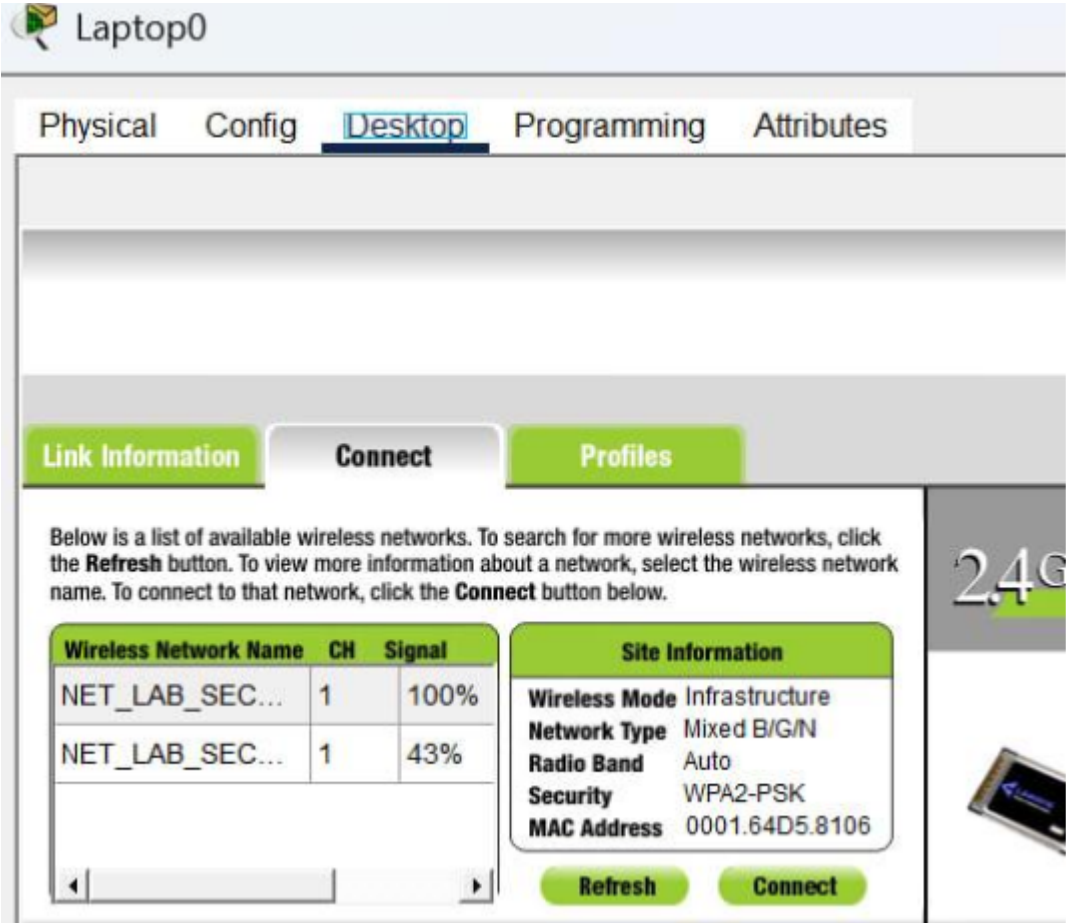
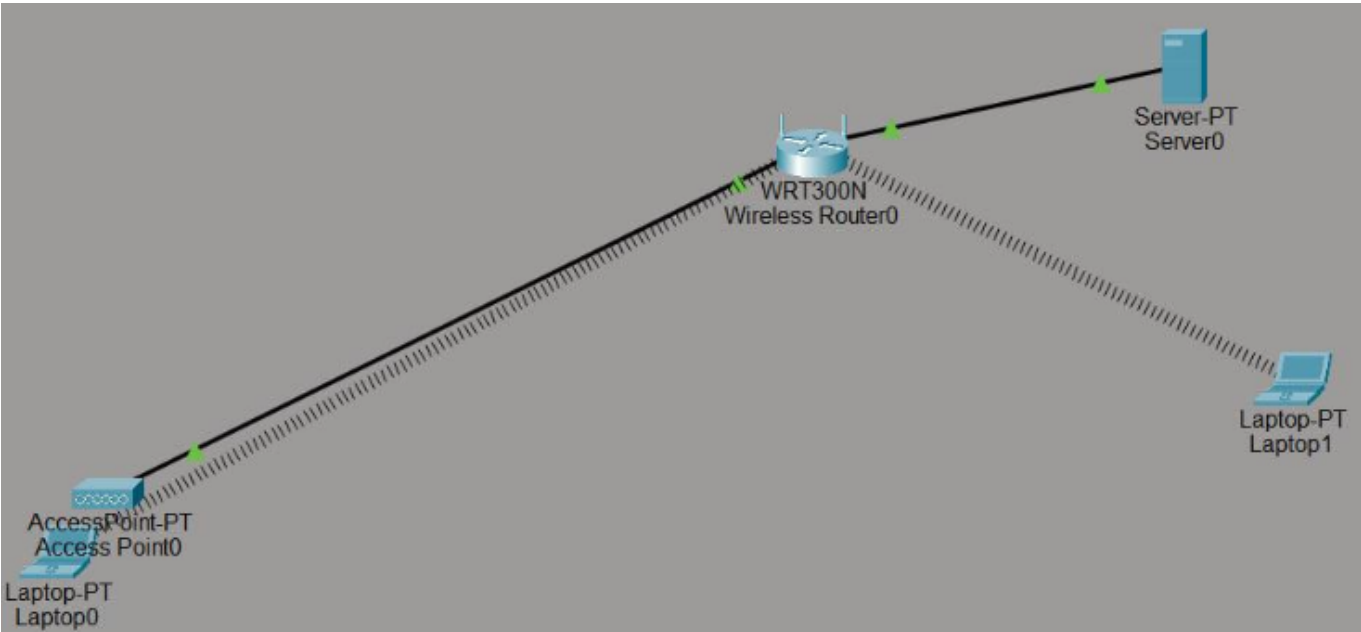
WEP Key

PSK Pass Phrase Wireless@Admin202

User ID

Password

Encryption Type AES



Part 4: Implementing Layer 2 Security


```
C:\>

C:\>ipconfig /all

Bluetooth Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0030.F2AB.ECDA
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                        0.0.0.0

    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-0E-56-64-39-00-D0-D3-70-6C-74
    DNS Servers.....: ::
                        0.0.0.0

Wireless0 Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 00D0.D370.6C74
    Link-local IPv6 Address.....: FE80::2D0:D3FF:FE70:6C74
--More--
```

Physical Config GUI Attributes

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless

Setup Wireless Security Access Restrictions Applications & Gaming Administration Status

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter Advanced Wireless Settings

Wireless MAC Filter

Wireless Port: 2.4G

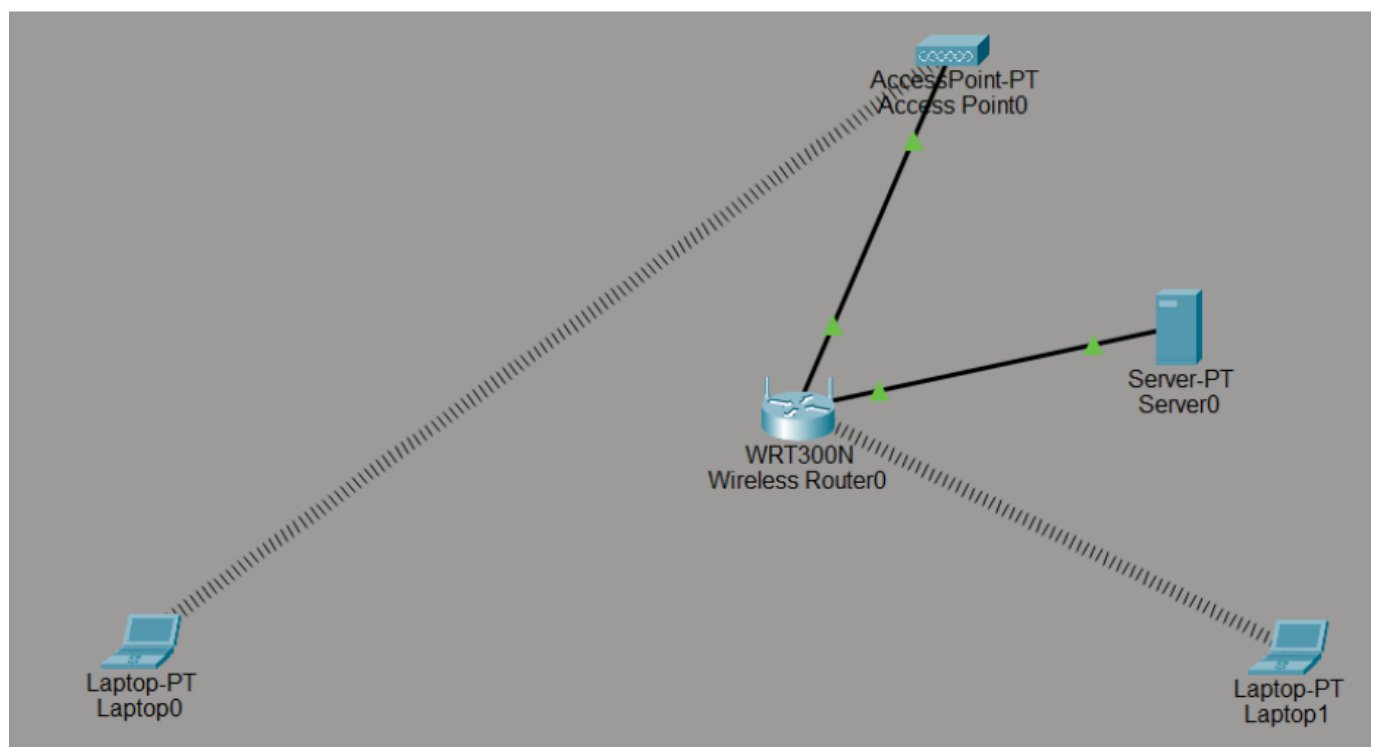
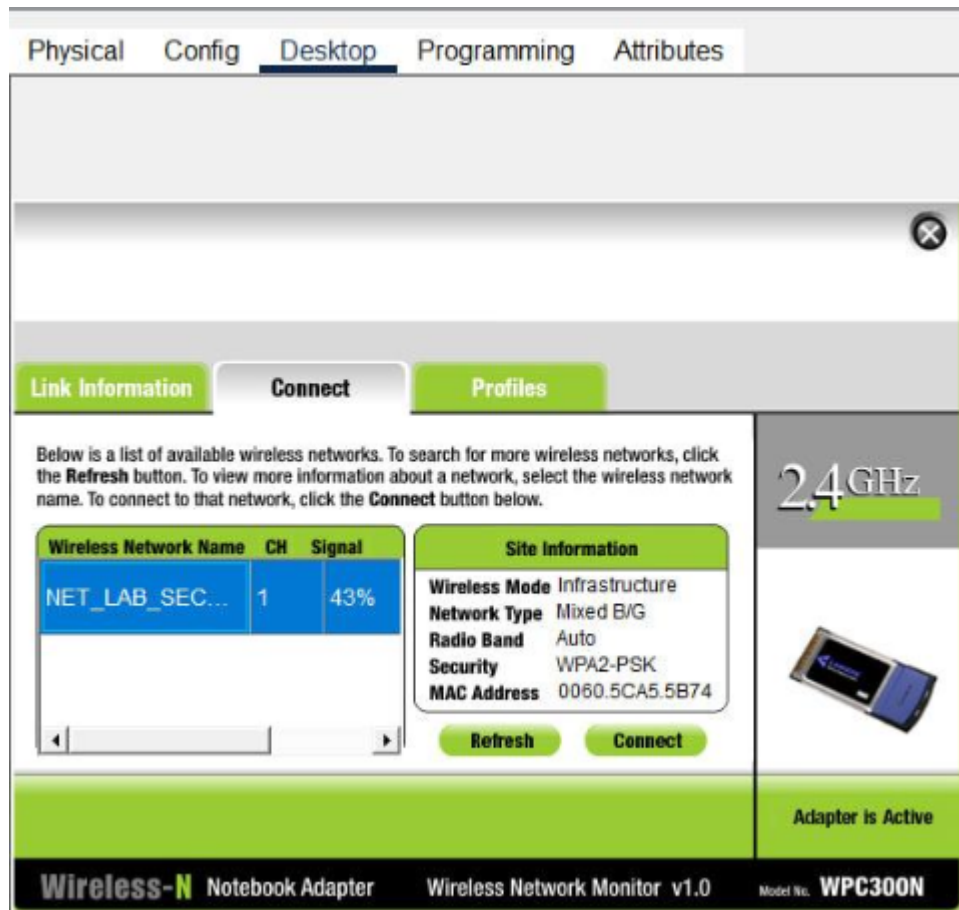
Access Resolution

MAC Address filter list

MAC 01: 30:F2:AB:EC:DA MAC 26: 0:00:00:00:00:00

MAC 02: 0:00:00:00:00:00 MAC 27: 0:00:00:00:00:00

Help...



Questions:

1. Why do we use non-overlapping channels like 1, 6, and 11 in professional deployments?
2. What happened to the connection when you applied the MAC filter? Is this an effective security measure on its own?
3. In Simulation Mode, what is the main difference you noticed between a "Beacon" frame and an "Association" frame?

Answers:

1. We don't use non-overlapping channels (1, 6, and 11) in professional deployments because it prevent interference between adjacent access points. 2.4GHz channels overlap, but these three have 25MHz separation so they don't interfere. Allows multiple APs in same area without performance degradation.
2. When you applied the MAC filter, the router blocked the connection from laptop0 immediately while Laptop1 worked normally. It's not an effective security measure on its own because it easily bypass MAC spoofing.
3. **Beacon:** * AP -> Broadcast, periodic advertisements * ongoing advertisements **Association:** * Client -> Specific AP, connection request * one-time connection setup.